

FIELD RESEARCH AND ONLINE RESEARCH AND DECISIONS

GROUND & ONLINE RESEARCH SUMMARY

Problem Analysis

1.1 Marine Decision-Making Landscape

Marine operations depend on **Satellite Earth Observation (EO)** products, **Potential Fishing Zone (PFZ)** advisories, weather forecasts, oceanographic observations, and geospatial datasets for activities such as fishing, navigation, coastal monitoring, and disaster response. Although organizations like **ISRO**, **INCOIS**, and **IMD** provide reliable marine information, these datasets are distributed across multiple independent platforms with different formats and update frequencies.

As a result, users must manually retrieve, interpret, and correlate information before making operational decisions. This fragmented workflow increases decision time, reduces operational efficiency, and limits the ability to respond quickly to changing marine conditions. The absence of an integrated **Marine Decision Intelligence Platform** capable of **spatial-temporal reasoning** highlights the need for an intelligent conversational system that delivers actionable, evidence-based recommendations.



Figure 1.1: Existing Marine Decision-Making Workflow

1.2 Problem Analysis

The study of the existing marine information ecosystem identified key challenges affecting decision-making:

- Fragmented access to marine information across multiple platforms.

- Heterogeneous oceanographic, meteorological, and geospatial datasets requiring manual correlation.
- Limited contextual reasoning and integrated decision support.
- Dispersed hazard advisories reducing situational awareness.
- Technical interfaces that are difficult for non-specialist users.
- Lack of **explainable AI** recommendations for operational planning.

These findings indicate that the primary challenge is not data availability but the absence of a unified platform capable of integrating heterogeneous datasets, performing contextual reasoning, and delivering intelligent conversational support.

1.3 Stakeholder Analysis

ORCA addresses the requirements of multiple marine stakeholders by providing domain-specific decision support.

Stakeholder	Primary Requirement	Expected Benefit
Fishermen	PFZ, weather, navigation	Improved safety and fishing efficiency
Maritime Operators	Route optimization, geofencing	Safer navigation
Researchers	Integrated EO and ocean data	Better marine analysis
Coastal Authorities	Boundary monitoring, alerts	Enhanced surveillance
Disaster Management	Early warnings	Faster emergency response
Environmental Organizations	Ecosystem monitoring	Sustainable resource management

Field Observation 1

The first respondent reported operating a **25-foot boat** and carrying approximately **10 litres of petrol/diesel per day**. The respondent stated that operations were conducted during daytime, primarily near the shore rather than in deep-sea conditions.

With respect to emergency management, the respondent stated that risks were generally undertaken without assistance from an application for instant event prediction. Sudden cyclones and lightning were identified as significant concerns, particularly because of the absence of advance preparation guidance.

The respondent also expressed a positive response to a potential voice-based application capable of providing assistance during emergencies, particularly where satellite communication could support emergency situations.

A requirement for a dedicated safe and shorter route incorporating **fuel analysis and weight analysis** was also positively received.

Field Observation 2

The second respondent stated that the available application provides required information and provides advance cyclone-related information, after which fishermen communicate among villages regarding whether to enter deep-sea areas.

Regarding communication between unfamiliar boats, the respondent indicated that direct interaction is difficult at distance and generally requires boats to approach one another. The interview notes further record that telephone communication was reported up to approximately **10 km**, while internet connectivity was reported to be poor in deep-sea conditions, with walkie-talkie or GPS-related communication being mentioned as alternatives.

The respondent reported regular use of GPS and also described the importance of local fisherman networks for receiving information and storm-related alerts. The notes further record the use of government-provided emergency equipment and emergency contact mechanisms during storm or cyclone situations.

For nearby-vessel awareness, the respondent stated that applications were not being used to identify nearby boats and that visual observation was used to detect nearby vessels and alter direction accordingly.

The interview notes also record location-based fishing information and the use of specific contact numbers associated with areas, along with a seasonal fishing period noted as **1 August to 31 May**.

1.7 Consolidated Field Findings

The on-ground analysis identified practical requirements relating to:

- Advance awareness of sudden weather hazards.
- Emergency communication and assistance.
- Reliable location and navigation support.
- Nearby-vessel awareness.
- Low-connectivity communication conditions.
- Fuel-related planning.
- Safe and shorter route identification.
- Accessible interaction, including voice-based assistance.



Figure 1.2: Stakeholder Ecosystem

1.4 Alignment with Smart India Hackathon Problem Statement 26176

ORCA directly addresses the requirements of **Problem Statement** through:

- Conversational natural language interaction.
- **Agentic AI** and **collaborative multi-agent architecture**.
- Autonomous planning and task decomposition.
- Integration of **Satellite Earth Observation**, oceanographic, meteorological, and GIS datasets.
- **Spatial-temporal reasoning** and **Geospatial Intelligence**.
- **Explainable, evidence-based recommendations**.

- Interactive maps, route optimization, and geofencing.
- Multilingual support for regional languages.

The proposed architecture aligns each functional module with the objectives of the problem statement, ensuring a comprehensive and intelligent marine decision-support platform.

1.5 Online Research and Existing Marine Information Ecosystem

The project research examined the marine information ecosystem provided by **ISRO, INCOIS, IMD, GIS providers and OpenStreetMap/public geospatial services**. The major information categories identified during the study were:

Source / Organisation	Information Studied
ISRO	Satellite imagery, Sea Surface Temperature (SST), Ocean Colour, Chlorophyll Concentration
INCOIS	Potential Fishing Zone (PFZ), Ocean State Forecast, wave height, currents
IMD	Weather forecasts, cyclone alerts, rainfall, wind speed, lightning advisories
GIS	Maritime boundaries, restricted zones, navigation layers, coastal mapping
OpenStreetMap / Public Geospatial Services	Base maps, coastal infrastructure and location services

In addition to secondary research, the team conducted direct discussions with fishermen to understand practical operating conditions, safety concerns, communication limitations and application requirements.

- Fuel-related planning.
- Safe and shorter route identification.
- Accessible interaction, including voice-based assistance.